

Reflection: RSLK Line Sensor: Arnav Krishnan

1. I learned about the physics behind the line sensors, and how they work by charging a capacitor and measuring how fast it discharges, with darker surfaces causing faster discharge. We capture this discharge over 1mc using Reflectance_Start() and Reflectance_End(). I was also able to see the actual applications of bitmasking.
2. The bitmasking approach to drive the RGB LED worked quite well, and it gave easy and intuitive (and visual) feedback about the robot's position without needing to a debugger/real time data collection. The LEDs made it easy for me (and Mr. Murray) to validate the correctness of our robot's behavior
3. I wonder why we don't use a weighted average of all 8 bits rather than just looking at bits 3 and 4. I imagine this could give much smoother line-following at best and would have no impact at worst.